

Blockchain Experiment 7

AIM: Implement the embedding wallet (Metamask) and transaction using Solidity

THEORY:

Q1: What is MetaMask?

MetaMask is a cryptocurrency wallet and browser extension that allows users to interact with blockchain networks such as Ethereum. It enables users to store, send, and receive digital assets securely. MetaMask also acts as a bridge between web browsers and decentralized applications (DApps), allowing users to sign transactions and deploy smart contracts. It provides account management, private key security, and seamless integration with platforms like Remix IDE.

Q2: What is a Test Network (Testnet)?

A test network, or testnet, is a blockchain environment used for testing purposes. It simulates the functionality of the main Ethereum network but uses virtual currency instead of real funds. Developers use testnets like Sepolia or Goerli to deploy and test smart contracts without financial risk. Testnets help identify bugs, verify functionality, and ensure proper execution before deploying applications on the mainnet.

Q3: Steps to Connect MetaMask with Remix IDE for Performing Transactions

- Install the MetaMask extension from the Chrome Web Store
- Create a new wallet or import an existing wallet using a secret recovery phrase
- Enable test networks in MetaMask settings
- Select a test network such as Sepolia
- Add test ETH using a faucet for performing transactions
- Open Remix IDE in the browser (<https://remix.ethereum.org>)
- Write and compile the smart contract in Remix
- Go to the Deploy & Run Transactions tab
- Select Injected Provider - MetaMask as the environment
- Connect MetaMask account when prompted
- Click Deploy and confirm the transaction in MetaMask
- Interact with deployed contract functions and approve transactions via MetaMask

CODE:

Smart Contract 1: OrderRecord.sol

Records each order placed on the platform permanently on the blockchain. Stores the buyer's wallet address, a comma-separated list of product IDs, the total amount in paisa, and a timestamp.

```
// SPDX-License-Identifier: MIT
```

```

pragma solidity ^0.8.0;
contract OrderRecord {
    struct Order {
        address buyer;
        string productIds;    // comma-separated e.g. "p1,p2,p3"
        uint256 totalPaisa;    // ₹199.99 stored as 19999
        uint256 timestamp;
    }

    Order[] public orders;

    event OrderPlaced(address buyer, string productIds, uint256
totalPaisa, uint256 timestamp);

    function placeOrder(string memory _productIds, uint256 _totalPaisa)
public {
        orders.push(Order(msg.sender, _productIds, _totalPaisa,
block.timestamp));
        emit OrderPlaced(msg.sender, _productIds, _totalPaisa,
block.timestamp);
    }

    function getOrderCount() public view returns (uint256) {
        return orders.length;
    }
}

```

Smart Contract 2: ReviewVerification.sol

Records product reviews on the blockchain to ensure authenticity and prevent tampering. Stores the product ID, reviewer's wallet address, star rating (1–5), and a timestamp. Also prevents the same wallet from reviewing the same product twice.

```

//Smart Contract 2: Review Verification.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract ReviewVerification {
    struct Review {
        string productId;

```

```

        address reviewer;
        uint8 rating;
        uint256 timestamp;
    }

    Review[] public reviews;
    mapping(string => mapping(address => bool)) public hasReviewed;

    event ReviewAdded(string productId, address reviewer, uint8 rating,
uint256 timestamp);

    function addReview(string memory _productId, uint8 _rating) public {
        require(_rating >= 1 && _rating <= 5, "Rating must be 1-5");
        require(!hasReviewed[_productId][msg.sender], "Already reviewed");

        reviews.push(Review(_productId, msg.sender, _rating,
block.timestamp));
        hasReviewed[_productId][msg.sender] = true;

        emit ReviewAdded(_productId, msg.sender, _rating,
block.timestamp);
    }

    function getReviewCount() public view returns (uint256) {
        return reviews.length;
    }
}

```

PROCEDURE

Step 1: Install Metamask Chrome extension

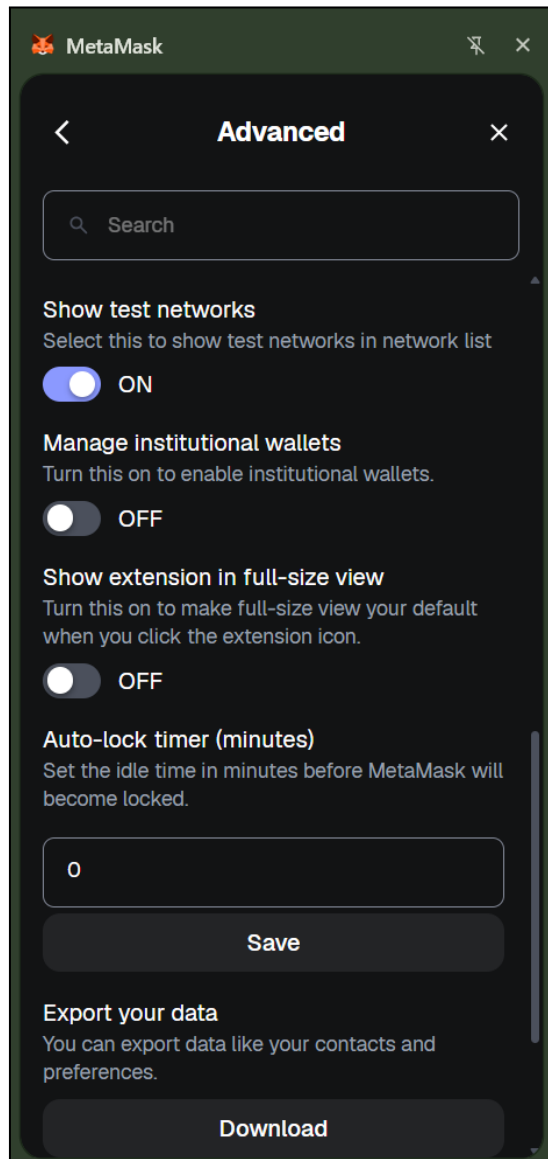
Install the MetaMask wallet extension from the Chrome Web Store. MetaMask is used to interact with the Ethereum blockchain and manage Ethereum accounts.

LINK:

<https://chromewebstore.google.com/detail/metamask/nkbihfbeogaeaoehlefnkodbefgpgknn?pli=1>

Step 2: Enable Test Network

Open MetaMask and enable Test Networks from the settings. Select the Sepolia Test Network which allows users to test smart contracts without using real cryptocurrency.



Step 3: Claim free test ETH

Visit the Sepolia Faucet website and request free test Ethereum (ETH). These tokens are used to deploy and test smart contracts on the Sepolia test network.

<https://sepolia-faucet.pk910.de/#!/claim/c21e4c9c-4717-417f-b7cb-7fb785414d23>

Sepolia PoW Faucet

Claim Rewards

Wallet: 0x5cE81fcFcD9c1F6fA512D4893323222AC53a3C84
Amount: 0.125 SepETH
Timeout: -

Claim Transaction has been confirmed in block #10469099!

TX: [0xc5fdacc1b458c1ecfb225d57b289bda53c5a573477b1ff3bc51ad9a59c66d572](https://etherscan.io/tx/0xc5fdacc1b458c1ecfb225d57b289bda53c5a573477b1ff3bc51ad9a59c66d572)

Did you like the faucet? Give that project a  Star 5,449

Or support this faucet by sharing your result with a



[Return to startpage](#)

Step 4: Open Remix IDE

Open the Remix Integrated Development Environment in the browser. Remix IDE is used to write, compile, deploy, and test Solidity smart contracts.

<https://remix.ethereum.org>

Step 5: Create Smart contracts

a. Order Record Contract

Records every order placed through the application permanently on the blockchain.

Stores on blockchain: buyer wallet address, product IDs (comma-separated), total amount in paisa, and timestamp.

- buyer wallet address
- product IDs (comma-separated)
- total amount in paisa
- timestamp.

Benefit:

- Prevents tampering of order records
- Provides transparent and immutable proof of purchase

b. Review Verification Contract

Ensures that product reviews are genuine and cannot be altered after submission.

Stores on blockchain:

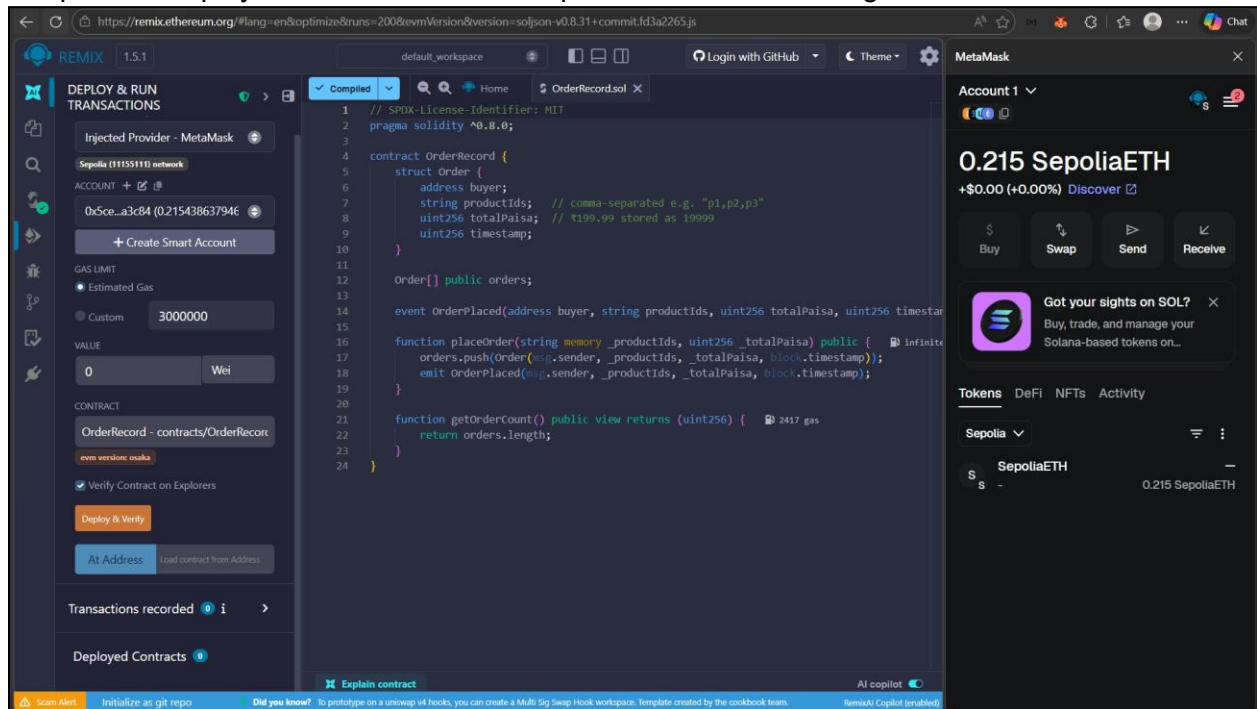
- product ID
- reviewer wallet address
- rating (1–5)
- timestamp.

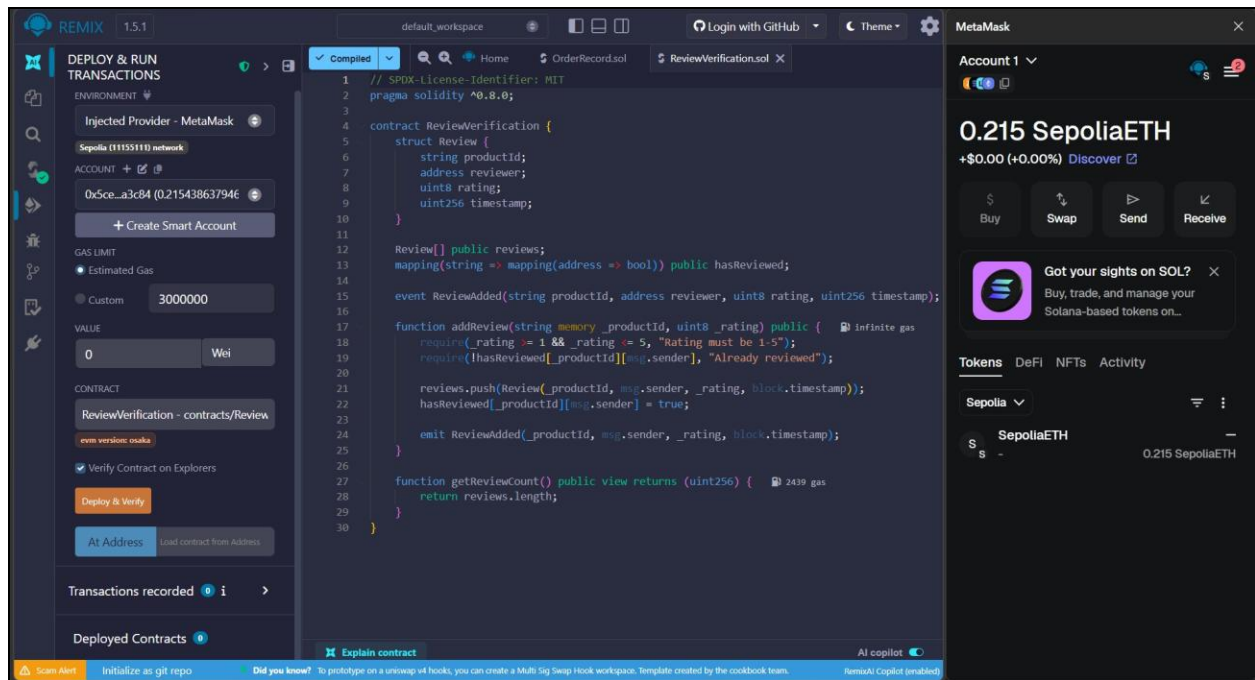
Benefit:

- Prevents fake or duplicate reviews from the same wallet
- Provides immutable, verifiable proof of each review

Step 6: Compile and deploy Smart contracts

Compile the smart contracts using the Solidity compiler in Remix IDE. After successful compilation, deploy the contracts on the Sepolia Test Network using the MetaMask wallet.

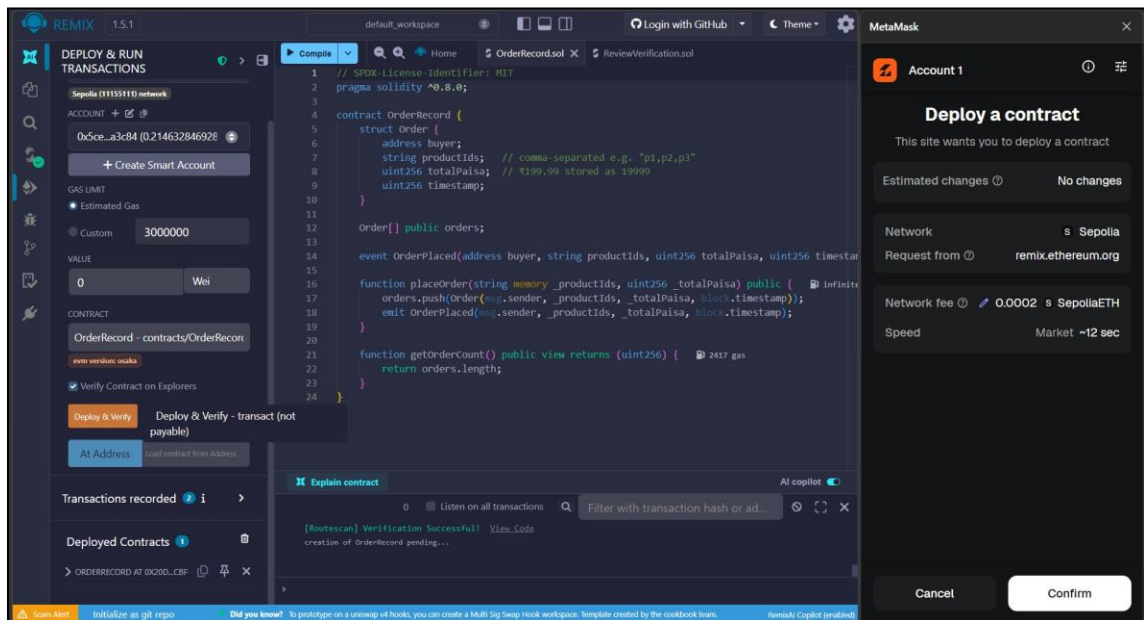




Step 7: Test the contracts

After deployment, test the contract functions such as `addProduct`, `getProduct`, and `addReview`. Verify that product details and reviews are stored correctly on the blockchain and can be retrieved using the contract functions.

a. Order Record Contract



localhost:5173

DI.Mart

Search for products...

0.2146 ETH

Cart

Hi, anupritamhapankar22

Logout

Dmart Products



Real Fruit Power Mixed Fruit
₹96

Add to Cart



B Natural Mixed Fruit Beverage
₹70

Add to Cart



Amul Taaza Toned Milk
₹71

- 1 +



Milky Mist Toned Milk
₹65

- 1 +



Thumbs Up
₹75

Add to Cart



Sprite
₹62

Add to Cart



Fortune Sunlite Sunflower Oil
₹150

Add to Cart



Gemini Refined Sunflower Oil
₹782

Add to Cart



Dakda



Milk



Milk



Milk

MetaMask

Account 1

0.215 SepoliaETH

+\$0.00 (+0.00%) Discover

Buy Swap Send Receive

Got your sights on SOL?

Buy, trade, and manage your Solana-based tokens on...

Tokens DeFi NFTs Activity

Sepolia

Mar 18, 2026

Contract deployment -0 SepoliaETH

Confirmed -0 SepoliaETH

localhost:5173/product/juice2

localhost:5173/cart

DI.Mart

Search for products...

0.2146 ETH

Cart

Hi, anupritamhapankar22

Logout

Shopping Cart



Amul Taaza Toned Milk
₹71

- 1 +

Remove

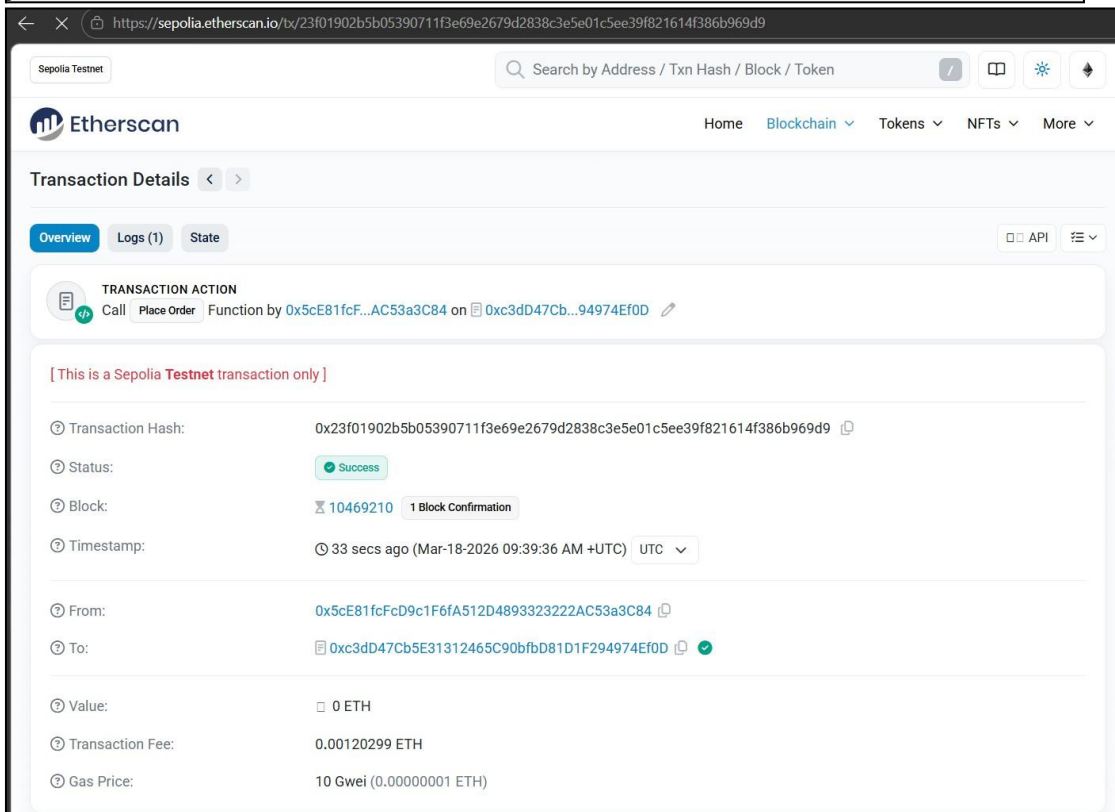
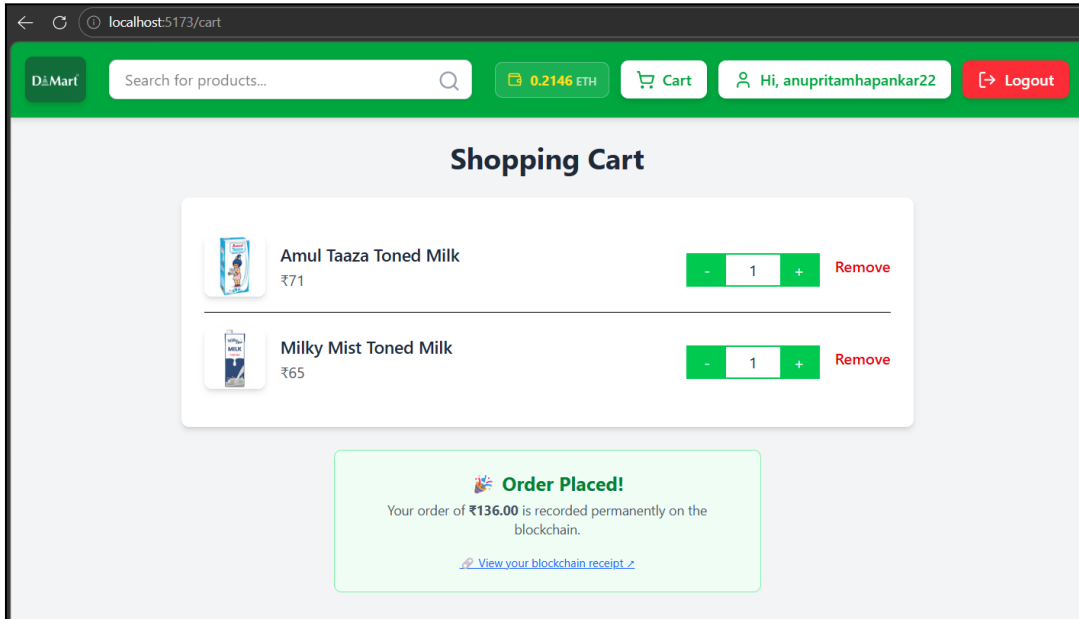


Milky Mist Toned Milk
₹65

- 1 +

Remove

Pay ₹136.00



b. Review Verification Contract

The image displays two screenshots related to a blockchain application. The top screenshot shows the Remix IDE interface with the `ReviewVerification.sol` contract being deployed. The contract code is visible in the center, and the deployment progress is shown in the bottom right. The bottom screenshot shows a web browser displaying a product page for "Real Mixed Fruit" with a price of ₹96. The page includes a search bar, a cart icon, and a "Write a Review" section. A MetaMask wallet is connected to the browser, showing a balance of 0.212 SepoliaETH.

Remix IDE - ReviewVerification.sol

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 contract ReviewVerification {
5     struct Review {
6         string productId;
7         address reviewer;
8         uint8 rating;
9         uint256 timestamp;
10    }
11
12    Review[] public reviews;
13    mapping(string => mapping(address => bool)) public hasReviewed;
14
15    event ReviewAdded(string productId, address reviewer, uint8 rating, uint256 timestamp);
16
17    function addReview(string memory _productId, uint8 _rating) public {
18        require(_rating >= 1 && _rating <= 5, "Rating must be 1-5");
19        require(!hasReviewed[_productId][_msg.sender], "Already reviewed");
20
21        reviews.push(Review(_productId, _msg.sender, _rating, block.timestamp));
22        hasReviewed[_productId][_msg.sender] = true;
23
24        emit ReviewAdded(_productId, _msg.sender, _rating, block.timestamp);
25    }
26
27    function getReviewCount() public view returns (uint256) {
28        return reviews.length;
29    }
30 }
```

Web Browser - Product Page

Search for products...

0.2123 ETH

Hi, anupritamhapankar22

Logout

Real Mixed Fruit

₹96

Available Quantity: 1 L

Add to Cart

Reviews

No reviews yet. Be the first to review this product!

Write a Review

Your Name

anupritamhapankar22

Rating: ★★★★★

Your Review

Great Product!

Submit Review

MetaMask

Account 1

0.212 SepoliaETH

+\$0.00 (+0.00%) Discover

Buy Swap Send Receive

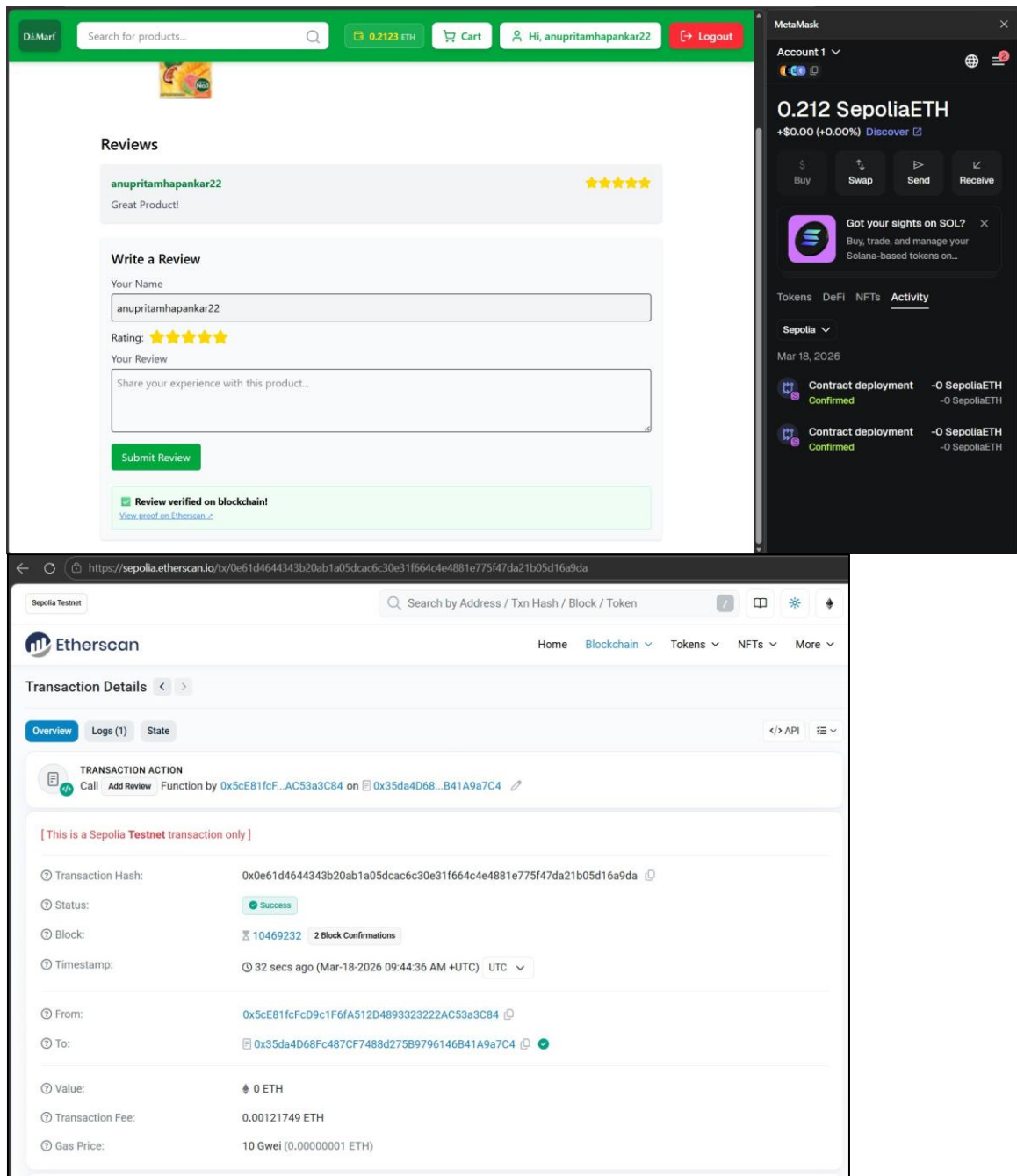
Got your sights on SOL? Buy, trade, and manage your Solana-based tokens on...

Tokens DeFi NFTs Activity

Mar 18, 2026

Contract deployment Confirmed -0 SepoliaETH

Contract deployment Confirmed -0 SepoliaETH



CONCLUSION:

Through this experiment, two smart contracts - OrderRecord and ReviewVerification, were successfully created and deployed on the Ethereum Sepolia test network using Solidity and Remix IDE. The experiment demonstrated how blockchain technology can be integrated into a full-stack web application to permanently and transparently record order transactions and product reviews. The use of the Sepolia test network and MetaMask allowed safe testing without real funds. Overall, this experiment provided practical understanding of designing, compiling, deploying, and testing smart contracts on the Ethereum blockchain.